



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
TRIAL EXAMINATIONS 2007

FORM VI

MATHEMATICS

Examination date

Wednesday 1st August 2007

Time allowed

3 hours (plus 5 minutes reading time)

Instructions

- All ten questions may be attempted.
- All ten questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the ten questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.
- Bundle the separate sheet with Question 10.

Checklist

- Folded A3 booklets: 10 per boy. A total of 1250 booklets should be sufficient.
- Candidature: 93 boys.

Examiner

SJE/LYL

QUESTION ONE (12 marks) Use a separate writing booklet.

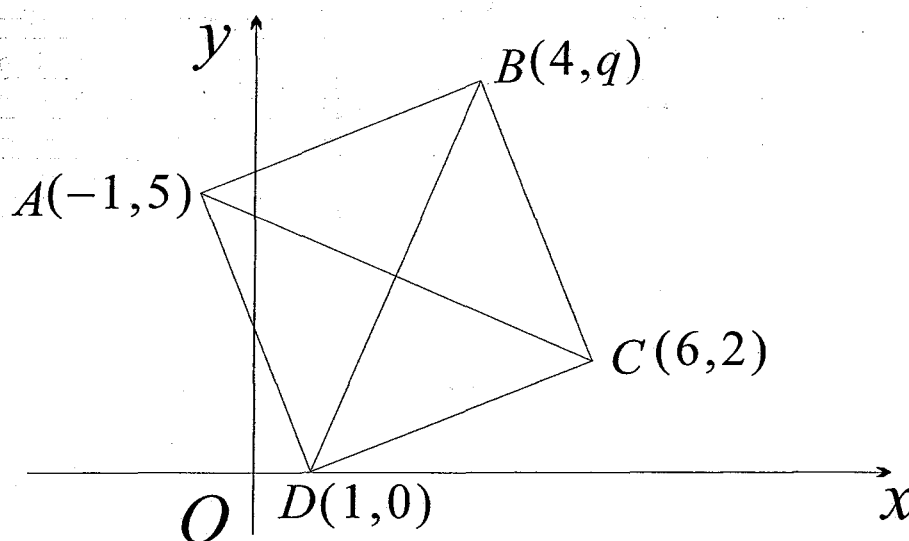
Marks

- (a) Evaluate $\sqrt[3]{43.1 - 16.98}$ correct to one decimal place. 2
- (b) Solve $x^2 = 3x$. 2
- (c) Solve $|x + 1| = 5$. 2
- (d) Differentiate $3x + \cos 2x$. 2
- (e) Write down the exact value of $\tan \frac{3\pi}{4}$. 1
- (f) Find a primitive function of $\frac{5}{x}$. 1
- (g) Find a and b if $a + b\sqrt{2} = (3 + \sqrt{2})(2 - \sqrt{2})$. 2

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

(a)



In the diagram above $ABCD$ is a square.

- (i) Find the gradient of AC . 1
- (ii) Show that the equation of BD is $7x - 3y - 7 = 0$. 2
- (iii) Find q , the y -coordinate of B . 1
- (iv) Find the length of AC . 1
- (v) Hence, or otherwise, find the area of $ABCD$. 1

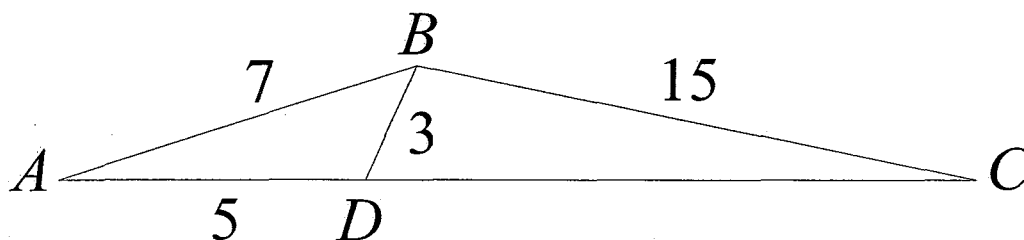
- (b) (i) Evaluate $\sum_{n=2}^4 \frac{n}{n+1}$. **1**
- (ii) Evaluate $\int_1^2 e \, dx$. **1**
- (c) Find the limiting sum of the geometric series $500 + 100 + 20 + 4 + \dots$. **2**
- (d) Find $\int \frac{x}{2x^2 + 3} \, dx$. **2**

QUESTION THREE (12 marks) Use a separate writing booklet.**Marks**

- (a) Differentiate with respect to x :
- (i) e^{x^2-9} **1**
- (ii) $x^2 \tan 5x$ **2**
- (b) Consider the parabola $(x-1)^2 = -6(y+4)$.
- (i) Write down coordinates of the vertex. **1**
- (ii) Find the equation of the directrix. **2**
- (c) Evaluate $\int_{\frac{\pi}{3}}^{\frac{3\pi}{2}} 2 \cos x \, dx$, leaving your answer as an exact value. **2**
- (d) Find the sum of the arithmetic series $1 + 4 + 7 + \dots + 226$. **2**
- (e) Find k such that $\int_1^k \frac{dx}{x} = 2$. **2**

QUESTION FOUR (12 marks) Use a separate writing booklet.**Marks**

(a)



In the diagram above, triangle ABC has dimensions $AB = 7$ cm and $BC = 15$ cm. The point D lies on AC such that $AD = 5$ cm and $BD = 3$ cm.

- (i) Use the cosine rule to show that $\angle ADB = 120^\circ$. 2
- (ii) Show that $\angle BCD = 10^\circ$ (rounded to the nearest degree). 2
- (iii) Find the length of DC , correct to the nearest millimetre. 2
- (b) The roots of the quadratic equation $px^2 - x + q = 0$ are -1 and 3 . Find p and q . 3
- (c) Find the equation of the normal to the curve $y = (2 - x)^3$ at the point where $x = 0$. 3

QUESTION FIVE (12 marks) Use a separate writing booklet.**Marks**

- (a) A particle moves in a straight line. At time t seconds, its displacement x metres from the origin is given by $x = 3 \cos \frac{t}{2}$, where $0 \leq t \leq 4\pi$.

- (i) Sketch the graph of x as a function of t . 2
- (ii) Find the times at which the particle is at rest. 2
- (iii) What is the particle's initial displacement and acceleration? 3
- (iv) Find the total distance travelled by the particle. 1

- (b) Consider the function $f(x) = \frac{x^2}{1 + x^2}$.

- (i) Show that $f''(x) = \frac{2(1 - 3x^2)}{(1 + x^2)^3}$. 3
- (ii) For what values of x is the function concave up? 1

QUESTION SIX (12 marks) Use a separate writing booklet.**Marks**

- (a) The table below shows the value of a function
- $f(x)$
- for three values of
- x
- .

2

x	3	4	5
$f(x)$	$\sqrt{7}$	$\sqrt{14}$	$\sqrt{23}$

Use the trapezoidal rule with the three given function values to find an approximation of $\int_3^5 f(x) dx$. Give your answer correct to one decimal place.

- (b) A ball is rolled up an inclined plane and is subject to an acceleration of $a = -6 \text{ m/s}^2$. Initially the ball has a velocity of $v = 12 \text{ m/s}$ and its displacement, measured from the bottom of the plane, is 36 m.

(i) Show that the velocity function is $v = 12 - 6t$.

1

(ii) Find the displacement as a function of time.

2

(iii) When does the ball reach the bottom of the plane and what is its speed then?

2

- (c) The fruit bat population in the Sydney Botanical Gardens has been increasing according to the equation $P = Ae^{kt}$, where A and k are constants. On 1st April 2005 there were 4800 bats, and by 1st April 2007 there were 10 800.

(i) Show that $k = \log_e \frac{3}{2}$.

2

(ii) If the trend continues without any intervention, how many bats will inhabit the gardens by 1st April 2010?

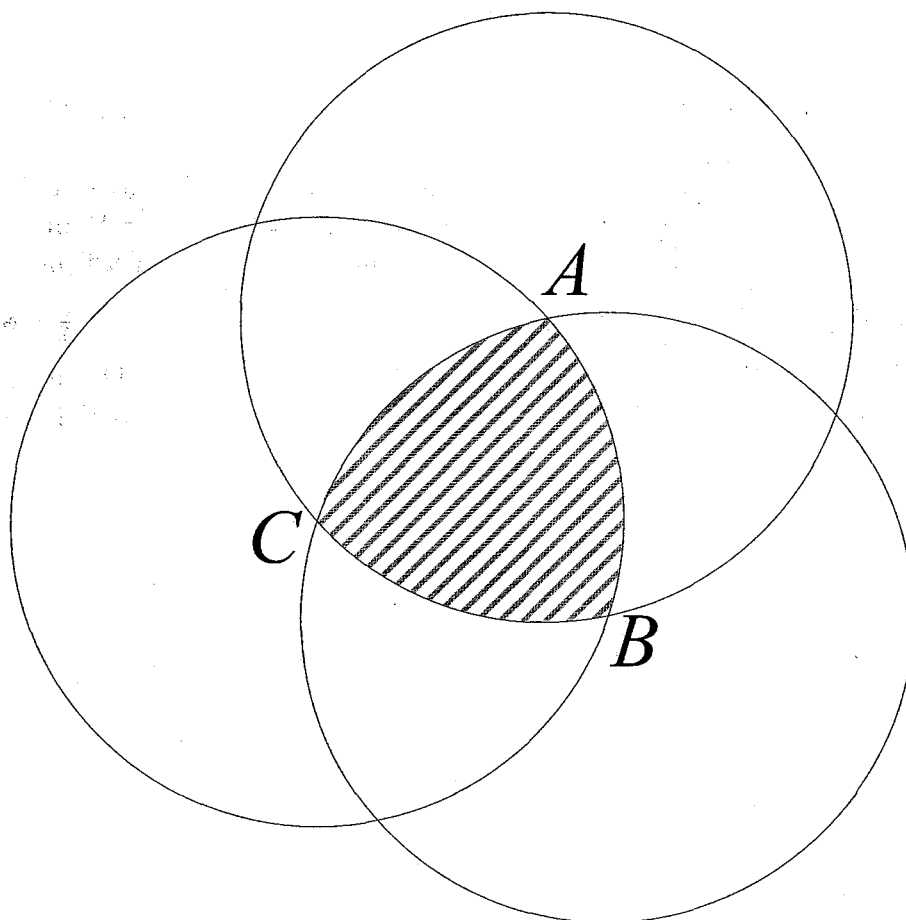
1

(iii) During what year is the bat population increasing at a rate of 6500 bats per year?

2

QUESTION SEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) Find the area between the curves $y = x^2$ and $x = y^2$. 3
- (b) Alex decides to invest \$30 000 into an investment fund offering 9% p.a. interest compounded monthly. How many months will it be before his money has doubled? Give your answer correct to the nearest month. 3
- (c) Gabriel's Horn is formed by rotating the area enclosed by the curve $y = \frac{1}{x}$ and the x -axis, between $x = 1$ and $x = a$, around the x -axis.
- (i) Find the volume of the horn when $a = 5$. 2
- (ii) Find the limiting value of the volume as a gets larger. 1
- (d)



In the diagram above, ABC is a Reuleaux Triangle. Its sides are equal arcs of congruent circles centred at A , B and C . The radius of each circle is 12 cm. Find:

- (i) the perimeter of the Reuleaux Triangle, 1
- (ii) the exact area of the Reuleaux Triangle. 2

QUESTION EIGHT (12 marks) Use a separate writing booklet.

Marks

- (a) Karen borrows \$15 000 from the bank. The loan plus interest and charges are to be repaid at the end of each month in equal monthly instalments of \$ M over 5 years. Interest is charged at 6% p.a. and is calculated on the balance owing at the beginning of each month. Furthermore, at the end of each month a bank charge of \$15 is added to the account.

Let A_n be the amount owing after n months.

- (i) Write down expressions for A_1 and A_2 and show that the amount owing after three months is given by

3

$$A_3 = 15\,000 \times 1.005^3 - (M - 15)(1 + 1.005 + 1.005^2).$$

- (ii) Hence write an expression of A_n .

1

- (iii) Find the monthly instalment, correct to the nearest cent.

3

- (b) The line $x - 4y + 2 = 0$ is a tangent to the parabola $x = Ay^2$, where A is a constant.

- (i) Form a quadratic equation and hence show that $A = 2$.

2

- (ii) Draw a neat sketch of the parabola and the tangent, showing the point of contact.

3

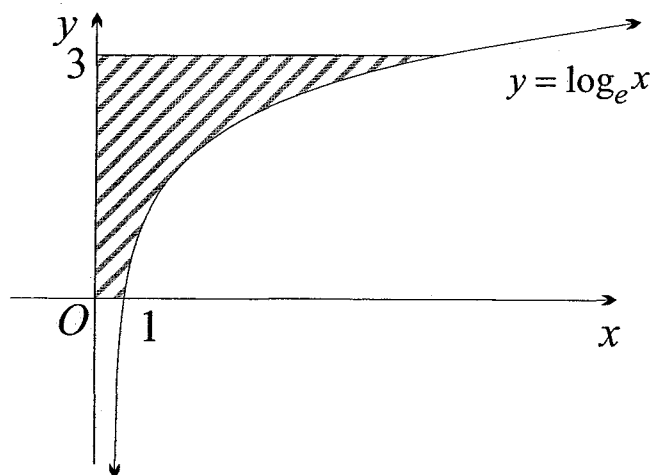
QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) (i) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$.

1

- (ii)

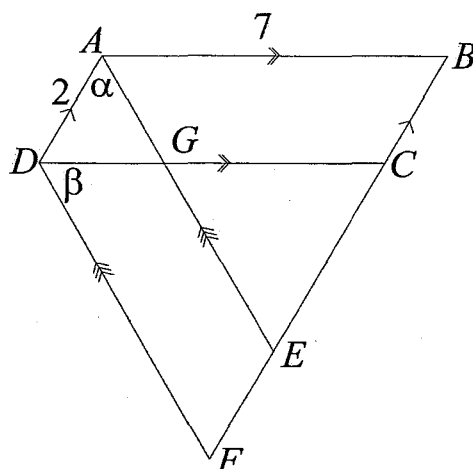


NOT TO
SCALE

The shaded region in the diagram above is bounded by the curve $y = \log_e x$, the line $y = 3$ and the coordinate axes. Using the result in part (i), or otherwise, find the exact area of the region.

2

(b)



In the diagram above, $ABCD$ and $AEFD$ are parallelograms. AE bisects $\angle DAB$ and DC bisects $\angle FDA$. Let $\angle DAG = \alpha$ and $\angle FDG = \beta$.

(i) Show that $\triangle AGD$ is equilateral, giving reasons.

3

(ii) If $AD = 2$ units and $AB = 7$ units, show that the area of the trapezium $ABFD$ is $\frac{77\sqrt{3}}{4}$ square units.

3

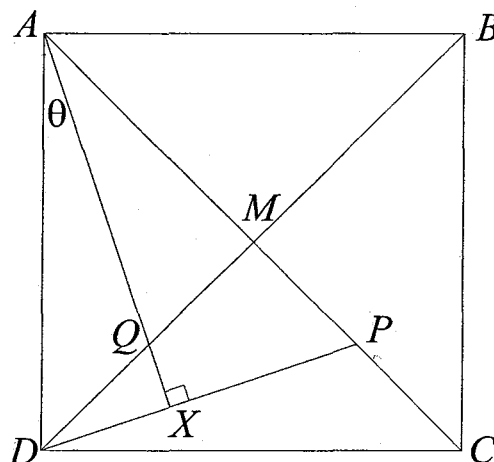
(c) Show that if $y = \frac{e^x + e^{-x}}{2}$ then $y'' = \sqrt{1 + (y')^2}$.

3

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

(a)



The square $ABCD$ is shown above. The diagonals AC and BD intersect at M . P is a point on the diagonal AC between M and C , and P is joined to D . The point X is chosen on DP so that $AX \perp DP$, and AX intersects the diagonal DB at Q . Let $\angle DAQ = \theta$.

The diagram has been reproduced on a separate sheet which can be used for your solution to this question. Insert this sheet with the rest of Question 10.

(i) Show that $\angle PDC = \theta$.

1

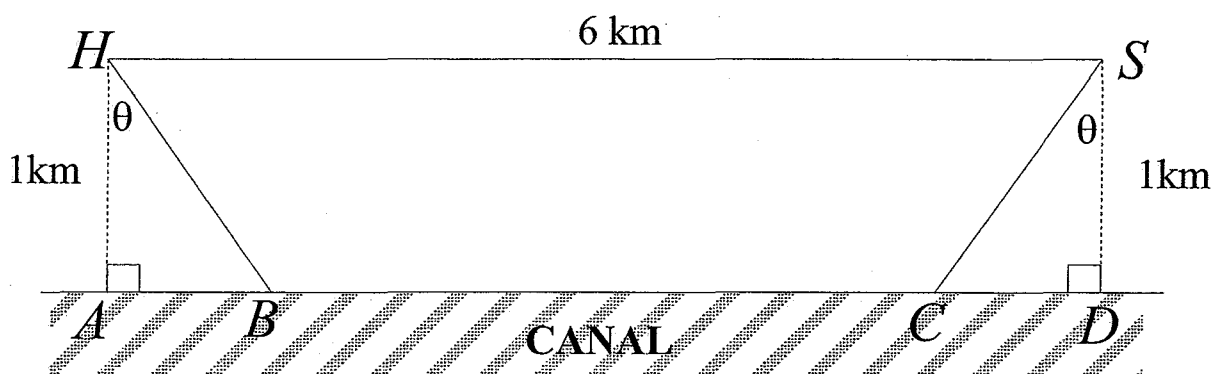
(ii) Hence show that $\triangle ADQ \cong \triangle DCP$.

3

(iii) Deduce that $\triangle DXQ \parallel \triangle AMQ$.

2

(b)

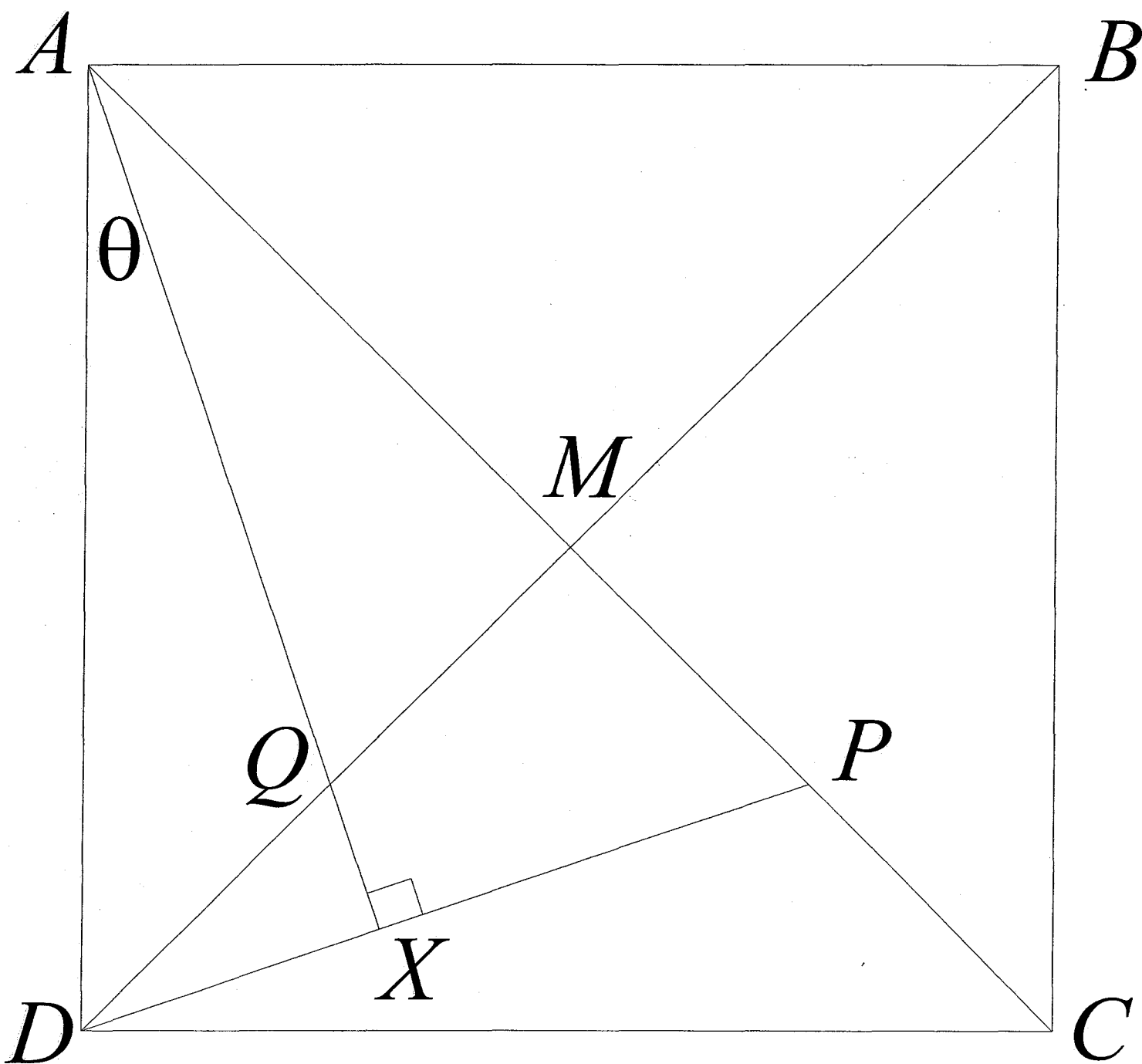


The diagram above shows that the distance between a boy's home H and his school S is 6 km. A canal $ABCD$ is 1 km from both his home and school. In winter the canal is frozen, so he take an alternate route $HBCS$, walking HB , skating BC and walking CS . His walking speed is 4 km/h and his skating speed is 12 km/h. Let $\angle AHB = \angle DSC = \theta$.

- (i) Show that the time taken for this alternate route is $T = \frac{1}{2 \cos \theta} + \frac{1}{2} - \frac{\tan \theta}{6}$. 2
- (ii) Find, to the nearest minute, the value of θ which minimises the time taken for the journey to school. 4

END OF EXAMINATION

Candidate Number



Q1

$$a) \sqrt[3]{26.12} \doteq 2.967 \quad (2.9 \text{ 2.96, etc } 1 \text{ mark}) \\ \doteq 3.0 \text{ (1dp)} \quad \checkmark \checkmark$$

$$b) x^2 = 3x$$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$$x = 0 \quad \checkmark \quad x - 3 = 0$$

$$x = 3 \quad \checkmark$$

$$c) |x+1| = 5$$

$$x+1 = 5 \quad \text{or} \quad x+1 = -5$$

$$x = 4 \quad \checkmark \quad \text{or} \quad x = -6 \quad \checkmark$$

$$d) 3 - 2 \sin 2x$$

$$e) \tan \frac{3\pi}{4} = -1 \quad \checkmark$$

$$f) 5 \log_e x + c \quad \checkmark$$

$$g) (3 + \sqrt{2})(2 - \sqrt{2}) \\ = 6 - 3\sqrt{2} + 2\sqrt{2} - 2 \\ = 4 - \sqrt{2} \quad \checkmark$$

$$a = 4 \quad b = -1 \quad \checkmark$$

Q2

$$a) i) \text{ gradient AC} = \frac{2-5}{6-1} \\ = -\frac{3}{5} \quad \checkmark$$

$$ii) \text{ gradient of BD} = \frac{7}{3} \quad \checkmark$$

equation of BD

$$\frac{y-0}{x-1} = \frac{7}{3} \quad \checkmark$$

$$y = \frac{7(x-1)}{3}$$

$$7x - 3y - 7 = 0$$

$$iii) \text{ Sub } x=4 \text{ into } 7x - 3y - 7 = 0$$

$$28 - 3y - 7 = 0$$

$$21 = 3y$$

$$y = 7 \quad \checkmark$$

$$iv) AC^2 = (6-1)^2 + (2-5)^2 \\ = 49 + 9 \\ = 58$$

$$AC = \sqrt{58} \text{ units} \quad \checkmark$$

$$v) \text{ Area ABCD} = \frac{1}{2} \times (\sqrt{58})^2 \\ = 29 \text{ square units} \quad \checkmark$$

$$b) i) \sum_{n=2}^4 \frac{n}{n+1} = \frac{2}{3} + \frac{3}{4} + \frac{4}{5} \\ = \frac{133}{60} \text{ or } 2\frac{13}{60} \quad \checkmark$$

$$ii) \int_1^2 e^x dx = [e^x]_1^2$$

$$= 2e - e$$

$$= e \quad \checkmark$$

$$c) a = 500 \quad \left. \begin{array}{l} r = \frac{1}{5} \end{array} \right\} \text{ either} \quad \checkmark$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{500}{1-\frac{1}{5}}$$

$$= 625 \quad \checkmark$$

$$d) \int \frac{x}{2x^2+3} dx = \frac{1}{4} \int \frac{4x}{2x^2+3} dx \quad \checkmark$$

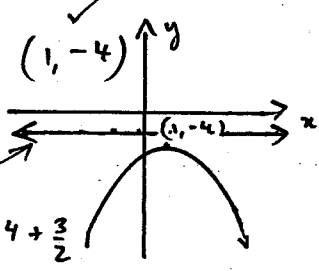
$$= \frac{1}{4} \log_e (2x^2+3) + C \quad \checkmark$$

Question 3

2) iii) $\frac{d}{dx}(e^{x^2-9}) = 2xe^{x^2-9}$ ✓

(ii) $\frac{d}{dx}(x^2 \tan 5x) = 2x \tan 5x + x^2 \cdot 5 \sec^2 5x$ ✓
 $= 2x \tan 5x + 5x^2 \sec^2 5x$ ✓

2) $(x-1)^2 = -6(y+4)$ ✓
i) Vertex is $(1, -4)$ ✓
ii) $4a = 6$ ✓
 $a = \frac{3}{2}$ ✓
directrix is $y = -4 + \frac{3}{2}$ ✓
 $y = -\frac{5}{2}$ ✓



2) $\int_{\pi/3}^{3\pi/2} 2 \cos x \, dx = \left[2 \sin x \right]_{\pi/3}^{3\pi/2}$ ✓
 $= -2 - 2 \frac{\sqrt{3}}{2}$ ✓
 $= -2 - \sqrt{3}$ ✓

2) $1 + 4 + 7 + \dots + 226$
 $a = 1$
 $d = 3$
 $T_n = a + (n-1)d$
So $226 = 1 + (n-1)3$
 $n-1 = 75$ ✓
 $n = 76$
Sum of series is $= \frac{n}{2}(a+l)$
 $= \frac{76}{2}(1+226)$
 $= 8626$ ✓

2) $\int_1^k \frac{dx}{x} = 2$
 $[\log_e x]_1^k = 2$
 $\log_e k - \log_e 1 = 2$ ✓
 $\log_e k = 2$
 $\therefore k = e^2$ ✓

$$a) i) \cos \angle ADB = \frac{5^2 + 3^2 - 7^2}{2 \times 5 \times 3} \quad \checkmark \checkmark$$

$$\angle ADB = 120^\circ$$

$$ii) \frac{\sin BCD}{3} = \frac{\sin 60}{15} \quad \checkmark$$

$$\sin BCD = \frac{3 \times \sin 60}{15} \quad \checkmark$$

$$BCD \doteq 10^\circ$$

(170° impossible as $170^\circ + 60^\circ > 180^\circ$)

$$iii) \frac{DC}{\sin 110^\circ} = \frac{3}{\sin 10^\circ} \quad \checkmark$$

$$DC = \frac{3 \sin 110^\circ}{\sin 10^\circ}$$

$$\doteq 16.2 \text{ cm} \quad \checkmark$$

$$b) -3 = \frac{q}{p} \quad \checkmark$$

$$2 = \frac{1}{p}$$

$$p = \frac{1}{2} \quad \checkmark$$

$$q = -\frac{3}{2} \quad \checkmark$$

$$c) y = (2-x)^3$$

$$y' = -3(2-x)^2 \quad \checkmark$$

$$m = -12$$

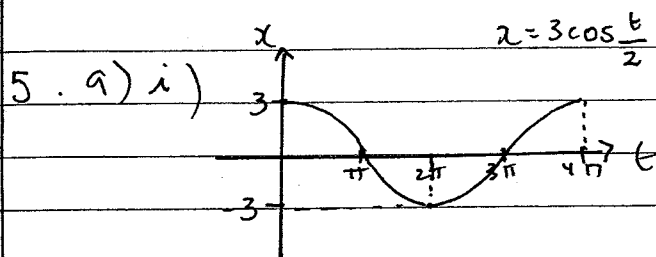
$$\text{normal } m = \frac{1}{12} \quad \checkmark$$

$$\text{when } x=0 \quad y=8$$

$$y-8 = \frac{1}{12}(x)$$

$$12y - 96 = x$$

$$x - 12y + 96 = 0 \quad \checkmark$$



5. a) i)

✓ correct shape, amplitude or period.

✓ remaining information correct and presented on the graph.

ii) At rest when $t=0$ ✓ one correct

$t=2\pi$ or $t=4\pi$ ✓ all three correct

iii) Initial displacement $x=3\text{m}$ ✓

$$x = 3 \cos \frac{t}{2}$$

$$\dot{x} = -\frac{3}{2} \sin \frac{t}{2}$$

$$\ddot{x} = -\frac{3}{4} \cos \frac{t}{2} \quad \checkmark$$

∴ at $t=0$ acceleration = $-\frac{3}{4} \text{ ms}^{-2}$ ✓

iv) Total distance travelled = 12m ✓

$$b) f(x) = \frac{x^2}{1+x^2}$$

$$f'(x) = \frac{(1+x^2)2x - x^2(2x)}{(1+x^2)^2} \quad \checkmark$$

$$= \frac{2x(1+x^2-x^2)}{(1+x^2)^2}$$

$$= \frac{2x}{(1+x^2)^2}$$

$$f''(x) = \frac{(1+x^2)^2 \cdot 2 - 2x \cdot 2 \cdot 2x(1+x^2)}{(1+x^2)^4} \quad \checkmark$$

$$= \frac{2(1+x^2)[(1+x^2)-4x^2]}{(1+x^2)^4}$$

$$= \frac{2(1-3x^2)}{(1+x^2)^3} \quad \checkmark$$

ii) Concave up when $f''(x) > 0$

$$1-3x^2 > 0$$

$$x^2 < \frac{1}{3} \quad \checkmark$$

$$-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$$

26

$$1) \int_a^b f(x) dx \equiv \frac{b-a}{2} (f(a) + f(b))$$

$$\begin{aligned} \int_3^5 f(x) dx &= \int_3^4 f(x) dx + \int_4^5 f(x) dx \\ &\equiv \frac{1}{2} (f(3) + f(4)) + \frac{1}{2} (f(4) + f(5)) \\ &\equiv \frac{1}{2} (\sqrt{7} + 2\sqrt{14} + \sqrt{23}) \\ &\equiv 7.5 \text{ (1dp)} \quad \checkmark \end{aligned}$$

$$b) \quad a = -6$$

$$i) \quad \frac{dx}{dt} = -6$$

$$x = -6t + C$$

$$t=0 \quad x=12$$

$$C = 12 \quad \checkmark$$

$$x = -6t + 12$$

$$ii) \quad \frac{dx}{dt} = -6t + 12$$

$$x = -\frac{6t^2}{2} + 12t + k \quad \checkmark$$

$$t=0 \quad x=36$$

$$k = 36 \quad \checkmark$$

$$x = -3t^2 + 12t + 36$$

$$iii) \quad x=0$$

$$-3t^2 + 12t + 36 = 0$$

$$3t^2 - 12t - 36 = 0$$

$$t^2 - 4t - 12 = 0$$

$$(t+2)(t-6) = 0$$

$$t \neq -2 \quad t = 6 \quad \checkmark$$

$$t = 6$$

$$x = 12 - 36$$

$$= -24 \text{ m/s}$$

$$\therefore \text{speed} = 24 \text{ m/s} \quad \checkmark$$

$$c) i) \quad P = Ae^{kt}$$

$$A = 4800$$

$$t = 2 \quad P = 10800$$

$$10800 = 4800 e^{2k} \quad \checkmark$$

$$\frac{108}{48} = e^{2k}$$

$$\frac{9}{4} = e^{2k}$$

$$\log \left(\frac{9}{4}\right)^2 = \log e^{2k} \quad \checkmark \text{ must show}$$

$$2 \log \frac{3}{2} = 2k$$

$$k = \log \frac{3}{2}$$

$$ii) \quad t = 5$$

$$P = 4800 e^{k5}$$

$$= 36450 \quad \checkmark$$

$$iii) \quad 6500 \text{ bats / year}$$

$$\frac{dP}{dt} = kP$$

$$6500 = kP$$

$$P = \frac{6500}{k}$$

$$P = Ae^{kt}$$

$$\frac{6500}{k} = 4800 e^{kt} \quad \checkmark$$

$$\frac{6500}{4800k} = e^{kt}$$

$$\log e \left(\frac{65}{48k}\right) = kt$$

$$t = \frac{\log e \left(\frac{65}{48k}\right)}{k}$$

$$\approx 2.97$$

$\therefore$ During 2008 the bat population is 6500 bats per year. $\checkmark$

$$t = 0 \quad 2005 \quad \text{(1st April)}$$

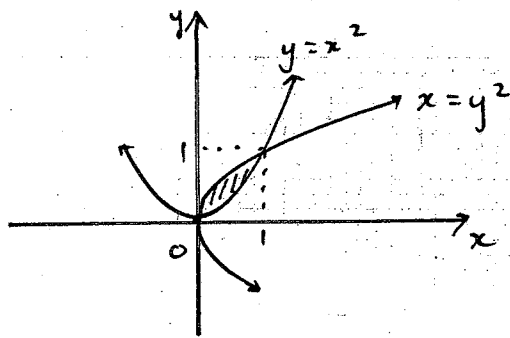
$$t = 1 \quad 2006 \quad \text{1st year}$$

$$t = 2 \quad 2007 \quad \text{2nd year}$$

$$t = 3 \quad 2008 \quad \text{3rd year}$$

Question 7

(a) Area = $\int_0^1 \sqrt{x} dx - \int_0^1 x^2 dx$ ✓
 $= \left[\frac{2}{3} x^{3/2} \right]_0^1 - \left[\frac{x^3}{3} \right]_0^1$ ✓
 $= \frac{2}{3} - \frac{1}{3}$ ✓
 $= \frac{1}{3}$ square unit. ✓



(b) $A = P(1+r)^n$
 $60000 = 30000 (1.0075)^n$ ✓
 $1.0075^n = 2$ ✓
 $n = \frac{\log_e 2}{\log_e 1.0075}$
 ≈ 92.77 ✓

$P = \$30\,000$

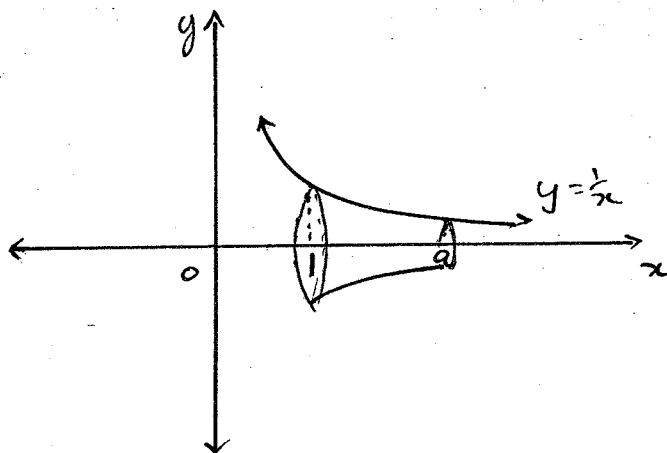
$r = 9\% \text{ p.a.}$

$= 0.0075$ ✓

$A = \$60\,000$ (doubled)

∴ His money has doubled after 93 months

(c) (i) $V = \pi \int_1^5 \left(\frac{1}{x}\right)^2 dx$
 $= \pi \int_1^5 x^{-2} dx$ ✓
 $= \pi \left[-x^{-1} \right]_1^5$
 $= \pi \left[-\frac{1}{5} - -1 \right]$ ✓
 $= \frac{4}{5} \pi$ cubic units

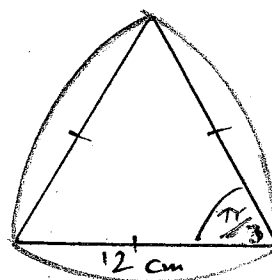


(ii) $V = \pi \left(-\frac{1}{a} - -1 \right)$
 $= \pi \left(1 - \frac{1}{a} \right)$

as $a \rightarrow \infty$, $V \rightarrow \pi$ ✓ ∴ limiting value is π cubic units.

(d) (i) Perimeter = $3 \times r\theta$
 $= 3 \times 12 \frac{\pi}{3}$
 $= 12\pi$ cm ✓

(ii) Area = Area Equilateral Triangle + 3 segments.
 $= \frac{1}{2} r^2 \sin \theta + 3 \left(\frac{1}{2} r^2 (\theta - \sin \theta) \right)$ ✓
 $= \frac{144}{2} \sin \frac{\pi}{3} + \frac{3}{2} \cdot 144 \cdot \frac{\pi}{3} - \frac{3}{2} (144) \sin \frac{\pi}{3}$
 $= 72 \frac{\sqrt{3}}{2} + 72\pi - 3(72) \frac{\sqrt{3}}{2}$
 $= 72\pi - 72\sqrt{3}$
 $= 72(\pi - \sqrt{3})$ cm² ✓



Question 8.

a) 6% p.a. = 0.005 per month

$$(i) A_1 = 15000(1.005) - M + 15$$

$$A_2 = 15000(1.005)^2 - M(1.005) + 15(1.005) - M + 15$$

$$\begin{aligned} A_3 &= 15000(1.005)^3 - M(1.005)^2 - M(1.005) - M + 15(1.005)^2 + 15(1.005) + 15 \\ &= 15000(1.005)^3 - M(1.005^2 + 1.005 + 1) + 15(1.005^2 + 1.005 + 1) \\ &= 15000(1.005)^3 - (M-15)(1.005^2 + 1.005 + 1) \\ &= 15000(1.005)^3 - (M-15)(1 + 1.005 + 1.005^2) \text{ as required.} \end{aligned}$$

$$(ii) A_n = 15000(1.005)^n - (M-15)(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$$

$$(iii) A_{60} = 0$$

$$\text{So } (M-15) \left[1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1} \right] = 15000(1.005)^{60}$$

$$\text{GP } a=1, r=1.005$$

$$\therefore (M-15) \left[\frac{1.005^{60} - 1}{1.005 - 1} \right] = 15000(1.005)^{60}$$

$$M = \frac{15000(1.005)^{60} \times 0.005}{1.005^{60} - 1} + 15$$

$$= \$304.99$$

b) (i) $x - 4y + 2 = 0$

$$x = 4y^2$$

$$4y^2 - 4y + 2 = 0$$

For tangent $\Delta = 0$

$$\text{So } (-4)^2 - 4A(2) = 0$$

$$16 - 8A = 0$$

$$A = 2$$

(ii) At point of intersection

$$2y^2 - 4y + 2 = 0$$

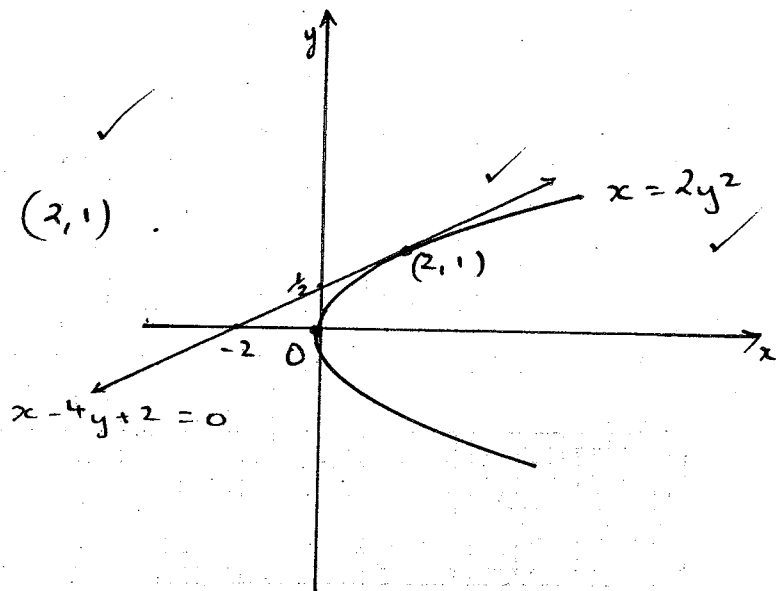
$$y^2 - 2y + 1 = 0$$

$$(y-1)^2 = 0$$

$$y = 1$$

$$\text{and } x = 2$$

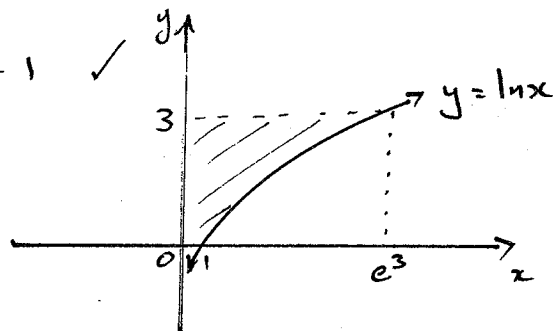
Coords. of point of intersection (2, 1)



Question 9

a) (i) $\frac{d}{dx} (x \ln x - x) = x \cdot \frac{1}{x} + \ln x - 1 \checkmark$
 $= \ln x$

(ii)
$$\begin{aligned} \text{Area} &= 3e^3 - \int_1^{e^3} \ln x \, dx \\ &= 3e^3 - [x \ln x - x]_1^{e^3} \checkmark \\ &= 3e^3 - [(e^3 \ln e^3 - e^3) - (0 - 1)] \\ &= 3e^3 - [3e^3 - e^3 + 1] \\ &= e^3 - 1 \text{ square units. } \checkmark \end{aligned}$$



Alternate approach:

$$\begin{aligned} \text{Area} &= \int_0^3 x \, dy \\ &= \int_0^3 e^y \, dy \\ &= [e^y]_0^3 \\ &= e^3 - e^0 \\ &= e^3 - 1 \text{ sq. units} \end{aligned}$$

b) (i) $\angle CAB = \alpha$
 $\angle ADG = \beta$

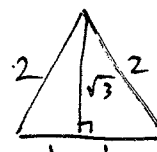
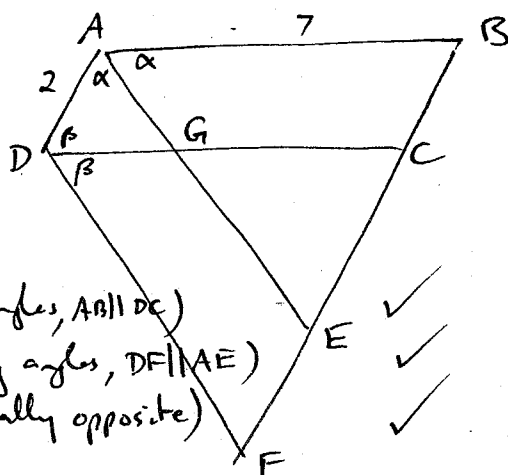
$\angle AGD = \alpha$ (alternate angles, $AB \parallel DC$)

$\angle AGC = \beta$ (corresponding angles, $DF \parallel AE$)

$\angle AGD = \angle CGE$ (vertically opposite)

$\therefore \alpha = \beta$

$\therefore \triangle ADG$ is equilateral.



(ii)
$$\begin{aligned} \text{Area} &= \text{Area } ABCD + \triangle DGF \\ &= 7\sqrt{3} + \frac{1}{2} 7^2 \sin 60^\circ \checkmark \\ &= 7\sqrt{3} + \frac{49}{2} \cdot \frac{\sqrt{3}}{2} \\ &= \frac{28\sqrt{3} + 49\sqrt{3}}{2} \\ &= \frac{77\sqrt{3}}{2} \text{ sq. units } \checkmark \end{aligned}$$

Height of parallelogram $ABCD = \sqrt{3}$

Alternate Method 1

$$\begin{aligned} \text{Area of trapezium} &= \frac{1}{2} h (a+b) \\ &= \frac{1}{2} \frac{7\sqrt{3}}{2} (2+7) \\ &= \frac{77\sqrt{3}}{4} \text{ sq. units} \end{aligned}$$

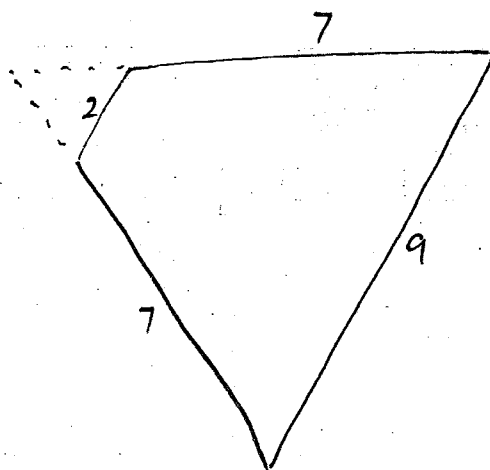
$h = \frac{7\sqrt{3}}{2}$

Question 9 (cont.)

Alternate Method 2

Consider area
of equilateral triangle
side lengths 9 units.

$$\begin{aligned} \text{Area} &= \frac{1}{2} 9^2 \sin 60^\circ - \frac{1}{2} 2^2 \sin 60^\circ \\ &= \frac{81}{2} \frac{\sqrt{3}}{2} - \frac{4}{2} \frac{\sqrt{3}}{2} \\ &= \frac{77\sqrt{3}}{4} \text{ sq. units.} \end{aligned}$$



(c) $y = \frac{e^x + e^{-x}}{2}$

$$y' = \frac{1}{2} (e^x - e^{-x})$$

$$y'' = \frac{1}{2} (e^x + e^{-x})$$

We need to show $y'' = \sqrt{1 + (y')^2}$

$$\text{LHS} = \frac{1}{2} (e^x + e^{-x})$$

$$\text{RHS} = \sqrt{1 + \left(\frac{1}{2}(e^x - e^{-x})\right)^2}$$

$$= \sqrt{1 + \frac{1}{4}(e^{2x} - 2 + e^{-2x})}$$

$$= \frac{1}{2} \sqrt{4 + e^{2x} - 2 + e^{-2x}}$$

$$= \frac{1}{2} \sqrt{e^{2x} + 2 + e^{-2x}}$$

$$= \frac{1}{2} \sqrt{(e^x + e^{-x})^2}$$

$$= \frac{1}{2} (e^x + e^{-x})$$

$$= \text{LHS}$$

Question 10

(a) (i) $\angle ADX = 180^\circ - 90^\circ - \theta$ (angle sum of $\triangle ADX$)
 $= 90^\circ - \theta$
 $\therefore \angle POC = 90^\circ - (90^\circ - \theta)$ (property of a square)
 $= \theta$ as required.

(ii) In triangles ADQ and DCP
 $AD = DC$ (sides of a square)
 $\angle PDC = \angle QAD = \theta$ (from (i) above)
 $\angle ADQ = \angle DCP = 45^\circ$ (diagonals of a square bisect vertex)
 $\therefore \triangle ADQ \equiv \triangle DCP$ (AAS)

(iii) In triangles DXQ and AMQ
 $\angle AMQ = 90^\circ$ (given)
 $\angle DXQ = 90^\circ$ (diagonals of a square intersect at right angles)
 $\angle AQM = \angle DQX$ (vertically opposite angles)
 $\therefore \triangle DXQ \parallel \triangle AMQ$ (AA)

b) (i) In $\triangle HAB$, $\cos \theta = \frac{1}{HB}$ and $\tan \theta = \frac{AB}{1}$
 $HB = \frac{1}{\cos \theta}$ $AB = \tan \theta$

Distance walked $= 2HB = \frac{2}{\cos \theta}$, $\therefore$ Time walked $= \frac{\frac{2}{\cos \theta}}{4}$ hours
Distance skated $= 6 - 2\tan \theta$, $\therefore$ Time skated $= \frac{6 - 2\tan \theta}{12}$ hours

So $T = \frac{1}{2\cos \theta} + \frac{6 - 2\tan \theta}{12}$
 $= \frac{1}{2\cos \theta} + \frac{1}{2} - \frac{\tan \theta}{6}$ as required.

(ii) $T = \frac{(\cos \theta)^{-1}}{2} + \frac{1}{2} - \frac{\tan \theta}{6}$

$\frac{dT}{d\theta} = \frac{-(-\sin \theta)(\cos \theta)^{-2}}{2} - \frac{\sec^2 \theta}{6}$
 $= \frac{\sin \theta}{2\cos^2 \theta} - \frac{1}{6\cos^2 \theta}$
 $= \frac{1}{\cos^2 \theta} (3\sin \theta - 1)$

For minimum $\frac{dT}{d\theta} = 0$

So

$3\sin \theta - 1 = 0$

$\sin \theta = \frac{1}{3}$

$\theta = 19^\circ 28'$

and

θ	10°	$19^\circ 28'$	30°
$\frac{dT}{d\theta}$	-0.94	0	$\frac{1}{9}$

$\therefore \theta = 19^\circ 28'$ is a minimum.

Test boundaries:

$\theta = 0$

$T = \frac{1}{4} + \frac{6}{12} + \frac{1}{4} = 1$ hour

$\theta = 90^\circ$

$T = \frac{6}{4} = 1\frac{1}{2}$ hours

and

$\theta = 19^\circ 28'$

$T = \frac{1}{2\cos 19^\circ 28'} + \frac{1}{2} - \frac{\tan 19^\circ 28'}{6}$

$= 58.28$ minutes.